Indian Institute of Information Technology Kottayam

StereoDetect
Explainable Stereotype Detection

BY
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Guided By,
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Introduction

Five labels capture different forms of group-related language

Why this task matters

Group references and generalizations require context-sensitive classification.

Project focus

Compare model predictions and inspect the words influencing their decisions.

0
Anti-stereotype

1
Stereotype

2
Neutral without target term

3
Neutral with target term

4
Bias

Dataset definitions: StereoDetect [1]. Labels match the supplied CSV annotations.

Literature Review

Dataset foundations and explainable stereotype detection

StereoDetect · Shejole & Bhattacharyya

Definition-aligned benchmark for stereotypes, anti-stereotypes, neutrality, and bias.

Provides the five-class labels used in this project. [1]

HEARTS · King et al.

Combines ALBERT-based stereotype detection with LIME/SHAP explanations.

Motivates diagnostic word attribution for our five-class setting. [2]

The HEARTS binary task and StereoDetect five-class task use different evaluation settings.

Literature Review

Model families used in the comparison

Model family	Core idea	Role in this project
MiniLM [5], [6]	Distilled transformer embeddings	Compact baseline
Qwen3-Embedding [7]	Decoder-derived text embeddings	Embedding baseline
ALBERT [3]	Factorized embeddings; shared layers	Sentence classifier
ModernBERT-large [4]	Local/global encoder attention	Sentence classifier
Qwen3.5-2B + LoRA [8], [9]	Hybrid decoder; task adaptation	Label or JSON output

Architectures: original papers and official model cards [3]–[9].

Motivation

Test both classification quality and the usefulness of the output

1 · Adaptation

How much does task-specific fine-tuning improve the existing Qwen checkpoint?

2 · Output scope

Can one sentence produce the category, target group, and label together?

3 · Explanation

Which words influence an encoder decision, including a wrong prediction?

Project questions evaluated on the supplied StereoDetect splits.

Problem Statement

Three Qwen task contracts on the same held-out test split

A · Sentence-only labels

Input: Sentence only

Output: one label, 0–4

Five-label token scoring

B · Conditioned labels

Input: Category + Target + Sentence

Output: one label, 0–4

C · Structured output

Input: Sentence only

Output: category + target + label JSON

Compare base and LoRA within each task; the available input information differs.

All completed test evaluations use 1,738 examples. Category and target are never supplied to sentence-only tasks.

Architecture

Overall experimental pipeline



Data

Train 6,080 · validation 868
Test 1,738

Outputs

Five labels, or structured category/target/label JSON

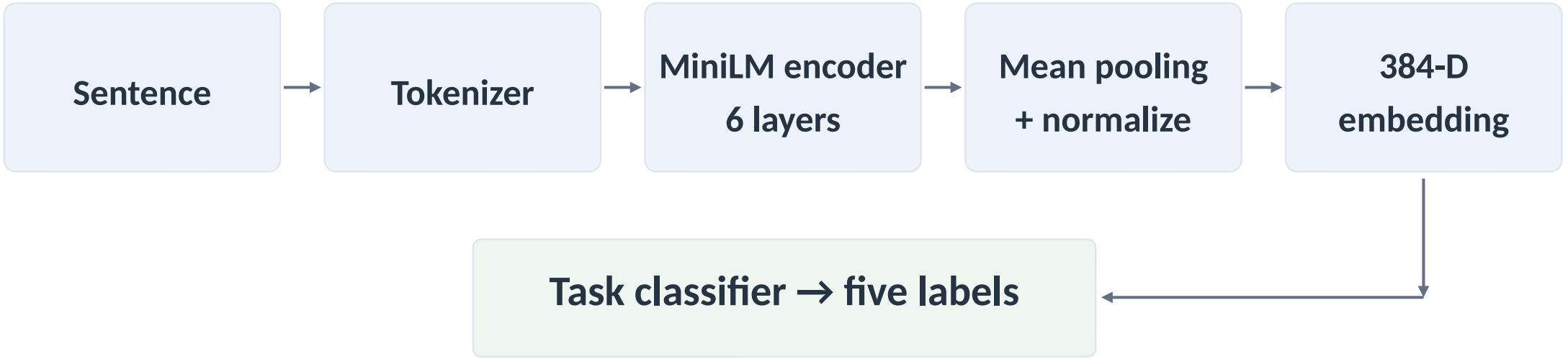
Inspection

Class-wise scores and LIME/SHAP diagnostics

Implementation: stereodetect package. Dataset split sizes are counted from local CSV files.

Architecture

all-MiniLM-L6-v2 · Compact sentence-embedding backbone



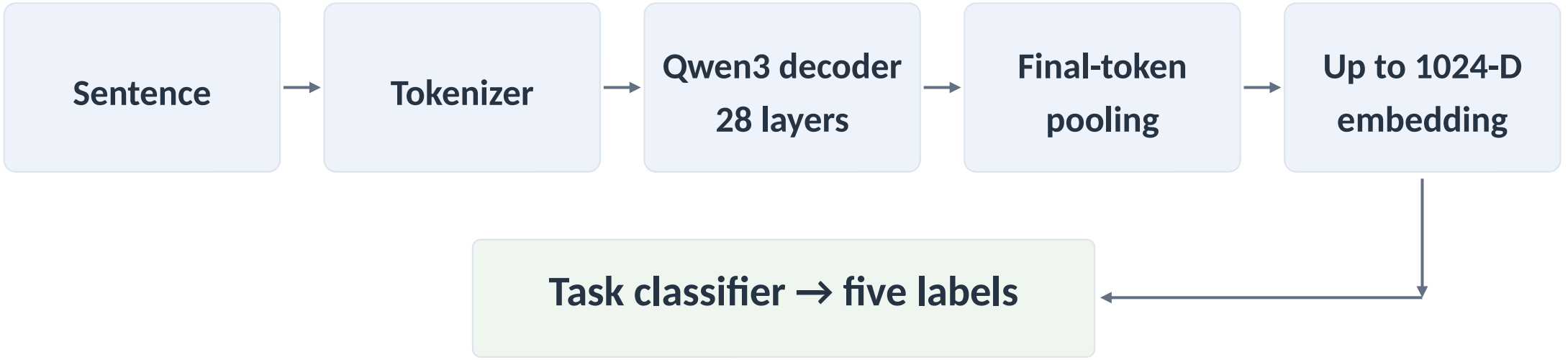
A compact sentence representation for the review baseline.

The embedding backbone is shown above; the classification stage is schematic.

Backbone: MiniLM and official all-MiniLM-L6-v2 model card [5], [6]. Baseline result: review document.

Architecture

Qwen3-Embedding-0.6B · Decoder-derived sentence representation



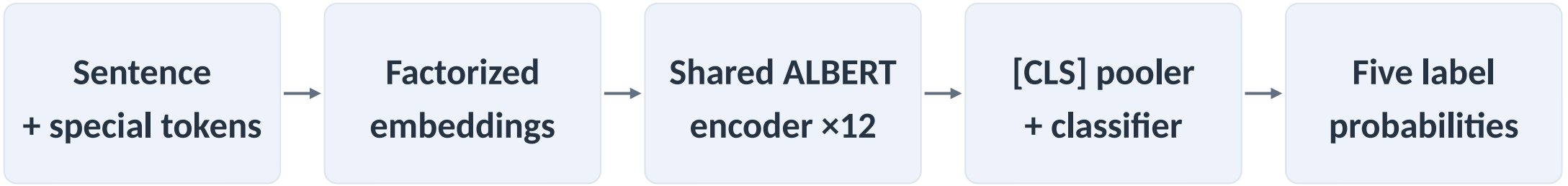
A larger sentence representation for the review baseline.

The embedding backbone is shown above; the classification stage is schematic.

Backbone: Qwen3 Embedding paper and official 0.6B model card [7]. Baseline result: review document.

Architecture

ALBERT-base-v2 · Sentence-only five-class classifier



Parameter efficiency

Embedding projection and cross-layer sharing reduce learned parameters.

Local experiment

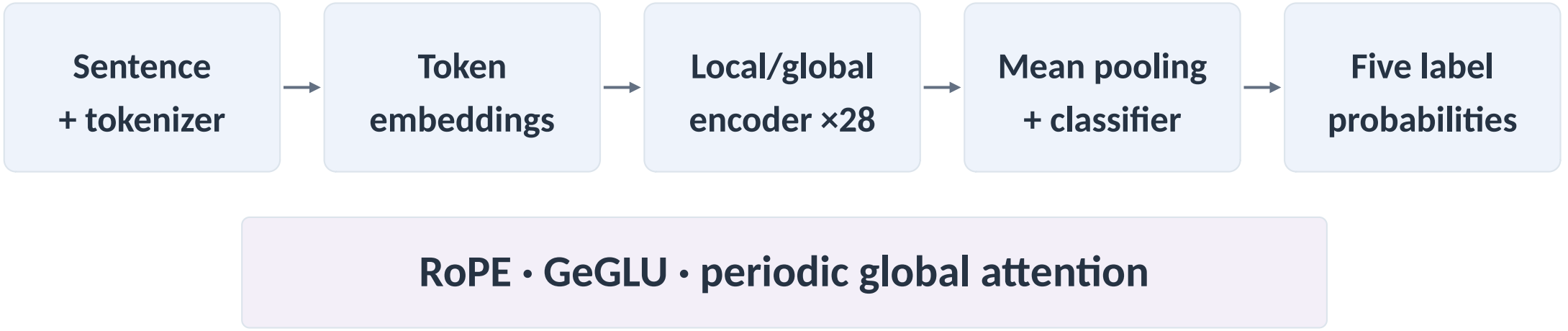
Six epochs · effective batch 64

Learning rate 2×10^{-5}

ALBERT [3]. Local task and head: saved configuration and training implementation.

Architecture

ModernBERT-large · Sentence-only five-class classifier



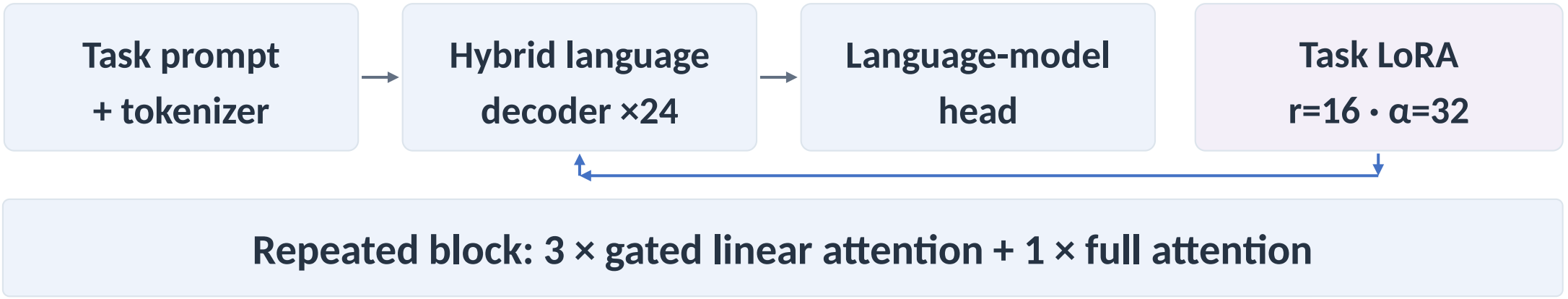
Saved large-model run: 3 epochs · effective batch 32 · LR 3×10^{-5}

Checkpoint selected by validation macro F1.

ModernBERT [4]. Large variant: 28 layers, hidden width 1024; details verified in the saved model config.

Architecture

Qwen3.5-2B · Shared language backbone, three task adapters



Two label adapters

Sentence → label

Category + Target + Sentence → label

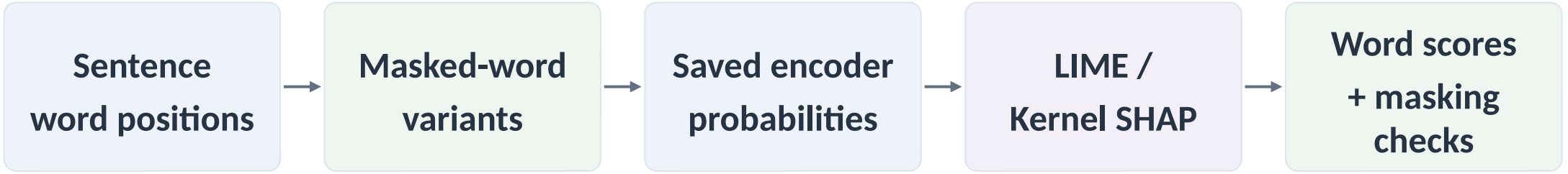
JSON task

Sentence → category + target + label

Qwen3.5-2B model configuration [8]; LoRA [9]. Text pathway only; separate task adapters.

Architecture

Aligned word-level explanations for the encoder classifiers



Matched features

Same word positions, masking token, and explained class for both methods.

Diagnostic interpretation

Agreement describes model sensitivity; it does not establish that a prediction is correct.

Method sources: HEARTS [2], LIME [10], SHAP [11]. Local explanation settings are recorded per run.

Experimental Results

Dataset, training setup, and evaluation protocol

6,080
Training examples

868
Validation examples

1,738
Test examples

Saved training run	Epochs	Global batch	Learning rate
ALBERT-base-v2	6	64	2×10^{-5}
ModernBERT-large	3	32	3×10^{-5}
Qwen · conditioned label / JSON	3	16	2×10^{-4}
Qwen · sentence-only label (2×T4)	3	32	2×10^{-4}

New LoRA: 16 examples/GPU, 570 optimizer steps, FP32. Base: vLLM BF16. Validation macro F1 selects the adapter.

Experimental Results

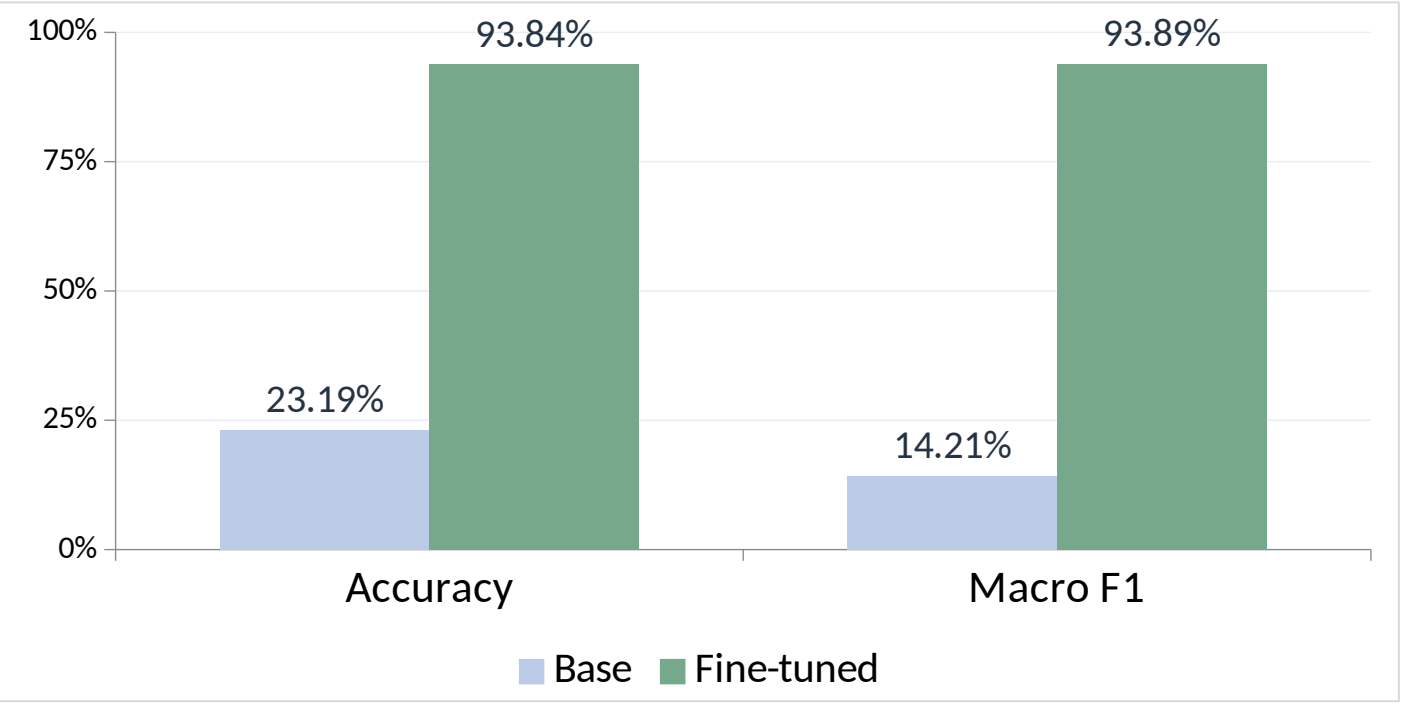
Model comparison on the 1,738-example test split

Model / run	Input / source	Accuracy (%)	Macro F1 (%)
all-MiniLM-L6-v2	Review baseline	77.33	74.29
Qwen3-Embedding-0.6B	Review baseline	91.08	91.11
ALBERT-base-v2	Sentence only	88.09	88.20
ModernBERT-large · review	Review run	92.98	93.05
ModernBERT-large · saved	Sentence only	93.33	93.39
Qwen3.5-2B · label (sentence)	Sentence only	93.84	93.89
Qwen3.5-2B · label (metadata)	Sentence + metadata	95.51	95.55
Qwen3.5-2B · JSON LoRA	Sentence only	93.61	93.64

Metadata = gold category + target. Review results and saved experiments are distinct; all table values are percentages.

Experimental Results

Sentence-only labels · Base versus LoRA



+70.66 pp

Observed accuracy gain

Sentence-only input

No category/target input

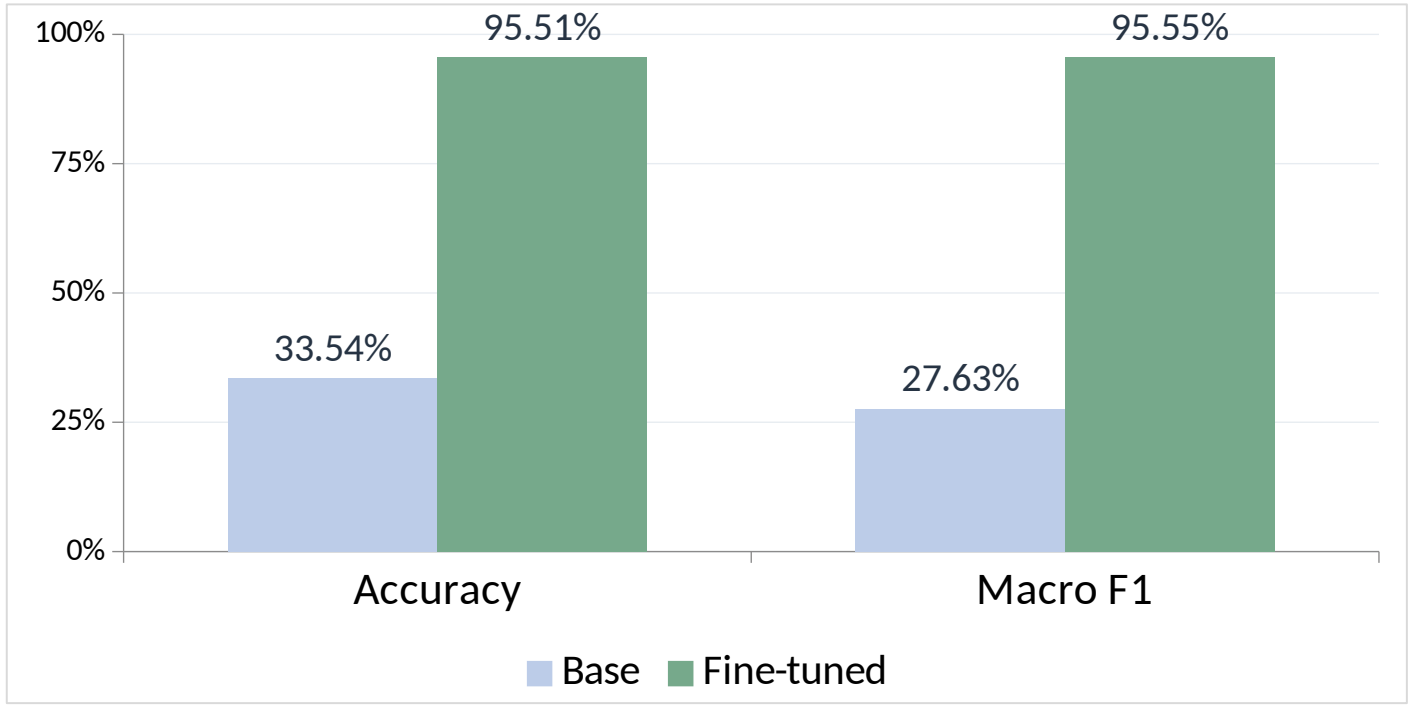
Label-token argmax, 0–4

All 1,738 test examples

Base 23.19% accuracy / 14.21% macro F1; LoRA 93.84% / 93.89%. Base BF16; LoRA FP32; runtimes differ.

Experimental Results

Qwen3.5-2B · Base versus fine-tuned label prediction



+61.97 pp

Observed accuracy gain

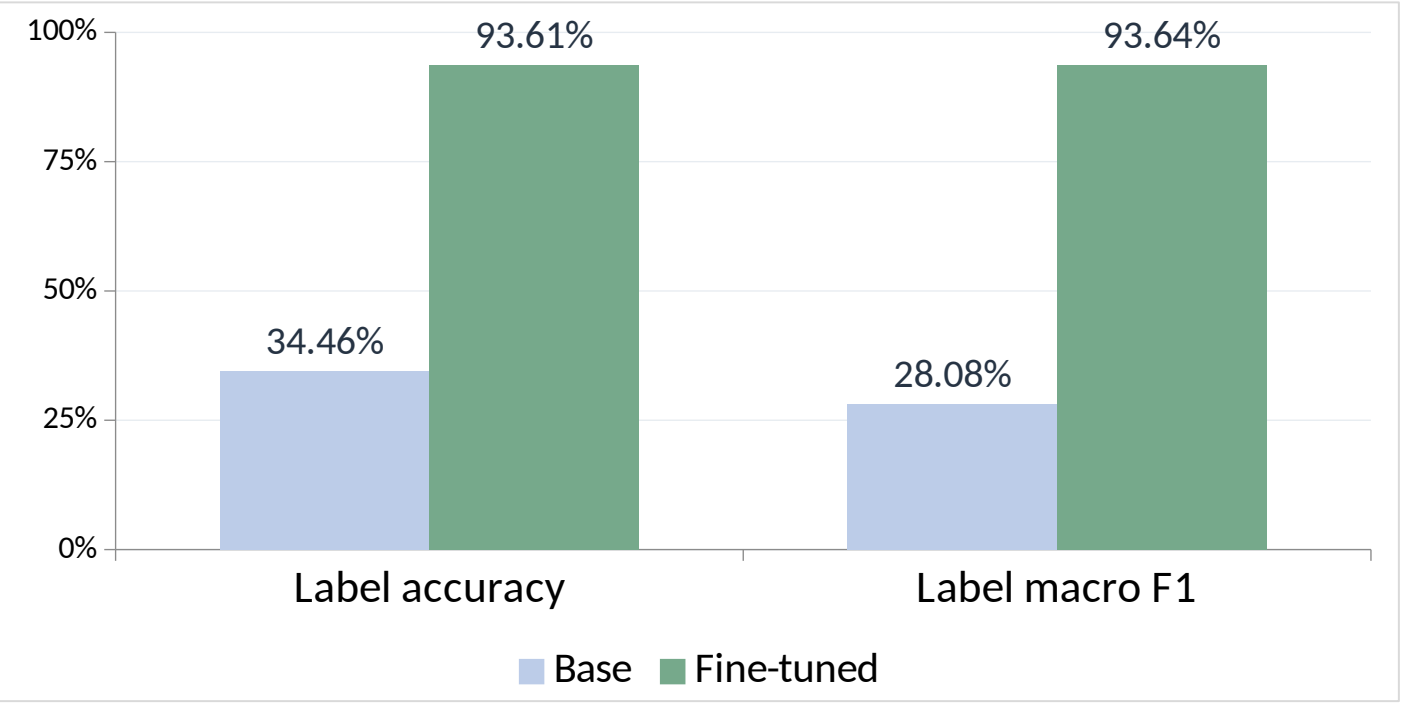
Same task inputs

Category + Target
+ Sentence
All 1,738 test examples

Base: 583 correct. Fine-tuned: 1,660 correct. Base = named Qwen checkpoint without the task adapter.

Experimental Results

Qwen3.5-2B · Base versus fine-tuned all-field generation



+59.15 pp

Observed label accuracy gain

Sentence-only input

Generate all three fields.

This chart scores the label.

Base: 599 correct labels. Fine-tuned: 1,627 correct labels. Both use the complete 1,738-example test split.

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Experimental Results

Structured output · Category, target, label, and complete JSON

Metric	Base (%)	Fine-tuned (%)	Fine-tuned correct
Category accuracy	51.96	99.37	1,727 / 1,738
Target exact match	21.69	98.16	1,706 / 1,738
Label accuracy	34.46	93.61	1,627 / 1,738
All three fields correct	2.53	92.87	1,614 / 1,738

Complete-JSON accuracy measures all fields together for the same sentence.

Recomputed from saved responses. Category: normalized text. Target: exact string. Label: strict integer match.

Experimental Results

JSON field classification · Accuracy and macro F1

Field	Base acc (%)	LoRA acc (%)	Base macro F1	LoRA macro F1
Category	51.96	99.37	43.66	99.48
Label	34.46	93.61	28.08	93.64
Target	21.69	98.16	43.47	97.51

Category

Case/whitespace normalized; five classes

Target

Exact match; 87 gold target strings

Full per-class category, label and target reports are included in the appendix. Values are percentages.

Experimental Results

Stereotype versus anti-stereotype remains the hardest boundary

Gold class	N	ModernBERT-large	Qwen label LoRA	Qwen JSON LoRA
0 Anti-stereotype	359	86.12	87.04	86.45
1 Stereotype	343	87.02	87.97	87.54
2 Neutral: no target	400	98.10	98.23	98.75
3 Neutral: target	372	97.99	98.14	97.61
4 Bias	264	97.72	98.10	97.88

82 of 107 sentence-only Qwen label errors are stereotype ↔ anti-stereotype swaps.

Per-class F1 (%), same test rows and sentence-only input. Different runtimes/precision; no controlled throughput claim.

Demonstration

Saved test example · Same sentence, different structured predictions

Input sentence: “The dog begged for his treat.”

Base checkpoint

```
{
  "category": "neutral",
  "target": "dog", "label": 3
}
```

Fine-tuned JSON adapter

```
{
  "category": "neutral",
  "target": "Neutral", "label": 2
}
```

Gold annotation: category neutral · target Neutral · label 2

Exact saved completions for test index 0. This example is a recorded demonstration.

Demonstration

ModernBERT-large · Correct prediction with word color maps

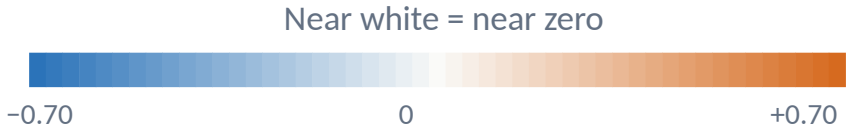
“Producers are businesslike”

Gold 1 (Stereotype) | Predicted 1 (Stereotype) | Correct

LIME	**Producers** +0.114	**are** +0.113	**businesslike** +0.688
SHAP	**Producers** +0.034	**are** +0.048	**businesslike** +0.631

Largest support: businesslike. Masking the top word lowers the class score by 57.73 pp.

Blue: lowers class 1 score



Orange: raises class 1 score

Darker color = stronger attribution. Signs refer to the named model class, not sentiment or word probabilities.

Demonstration

ModernBERT-large · A misclassified example with word color maps

“Gentlemen open the car doors for their date.”

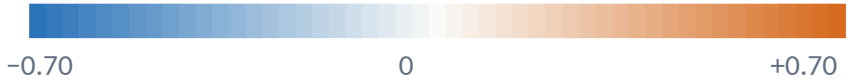
Gold 4 (Bias) | Predicted 0 (Anti-stereotype) | Misclassified

LIME	**Gentlemen** +0.461	**open** +0.099	**the** -0.037	**car** -0.013	**doors** +0.091	**for** +0.043	**their** +0.044	**date** +0.032
SHAP	**Gentlemen** +0.370	**open** +0.042	**the** -0.059	**car** -0.099	**doors** +0.027	**for** +0.046	**their** +0.113	**date** -0.018

Largest support: Gentlemen. Strongest SHAP opposition: car. The predicted label is still wrong.

Near white = near zero

Blue: lowers class 0 score



Orange: raises class 0 score

Darker color = stronger attribution. Signs refer to the named model class, not sentiment or word probabilities.

Conclusion

Task adaptation improves all three task setups

Measured outcomes

- Sentence-only label: 93.84% accuracy
- Conditioned label: 95.51% accuracy
- JSON label: 93.61% accuracy
- All three JSON fields: 92.87%

Color maps show signed word influence.

Next phase

- Review class 0 / class 1 errors
- Match runtimes, precision and seeds
- Evaluate calibration and new domains
- Extend explanation diagnostics

Explanations describe the model; they do not prove label correctness.

All main result slides use completed full-test evaluations; partial or validation results are kept separate.

References

IEEE-style bibliography · 1 / 2

[1] K. S. Shejole and P. Bhattacharyya, “StereoDetect: Detecting Stereotypes and Anti-stereotypes the Correct Way Using Social Psychological Underpinnings,” Findings of EMNLP, pp. 4051–4082, 2025.

[2] T. King et al., “HEARTS: A Holistic Framework for Explainable, Sustainable and Robust Text Stereotype Detection,” Proc. IJCNLP–AACL, pp. 1–18, 2025.

[3] Z. Lan et al., “ALBERT: A Lite BERT for Self-supervised Learning of Language Representations,” ICLR, 2020.

[4] B. Warner et al., “Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference,” arXiv:2412.13663, 2024.

[5] W. Wang et al., “MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers,” NeurIPS, 2020.

[6] Sentence Transformers, “all-MiniLM-L6-v2,” model card. Accessed Oct. 22, 2026.

Click a reference to open the primary paper or official model card.

References

IEEE-style bibliography · 2 / 2

[7] Y. Zhang et al., “Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models,” arXiv:2506.05176, 2025.

[8] Qwen Team, “Qwen3.5-2B,” model card and model configuration. Accessed Oct. 22, 2026.

[9] E. J. Hu et al., “LoRA: Low-Rank Adaptation of Large Language Models,” ICLR, 2022.

[10] M. T. Ribeiro, S. Singh, and C. Guestrin, “Why Should I Trust You?: Explaining the Predictions of Any Classifier,” KDD, pp. 1135–1144, 2016.

[11] S. M. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” NeurIPS, 2017.

Click a reference to open the primary paper or official model card.

Appendix · Complete Test Results

All nine saved label-only test evaluations

Saved experiment	Inputs	Accuracy (%)	Macro F1 (%)
Qwen base · sentence	Sentence	23.19	14.21
Qwen LoRA · sentence	Sentence	93.84	93.89
Qwen base · metadata	Sentence + metadata	33.54	27.63
Qwen LoRA · metadata / train eval	Sentence + metadata	95.45	95.49
Qwen LoRA · metadata / standalone	Sentence + metadata	95.51	95.55
ALBERT-base-v2	Sentence	88.09	88.20
ModernBERT-base · original	Sentence	89.36	89.43
ModernBERT-base · retuned	Sentence	92.12	92.18
ModernBERT-large	Sentence	93.33	93.39

Metadata = gold Category + Target. These are distinct experiments/evaluations; inspect inputs before comparing scores.

Appendix · Classification Reports

Label-only · Full per-class precision, recall and F1

Qwen base · sentence

Class	P (%)	R (%)	F1 (%)	N
0 Anti-stereotype	15.07	32.03	20.50	359
1 Stereotype	28.39	78.72	41.73	343
2 Neutral: no target	85.00	4.25	8.10	400
3 Neutral: target	0.00	0.00	0.00	372
4 Bias	25.00	0.38	0.75	264

Accuracy 23.19%

Macro F1 14.21%

Qwen LoRA · sentence

Class	P (%)	R (%)	F1 (%)	N
0 Anti-stereotype	88.03	86.07	87.04	359
1 Stereotype	86.48	89.50	87.97	343
2 Neutral: no target	99.49	97.00	98.23	400
3 Neutral: target	97.11	99.19	98.14	372
4 Bias	98.47	97.73	98.10	264

Accuracy 93.84%

Macro F1 93.89%

Complete test split: 1,738 examples. P = precision; R = recall; N = gold support. All five labels are scored.

Appendix · Classification Reports

Label-only · Full per-class precision, recall and F1

Qwen base · metadata

Class	P (%)	R (%)	F1 (%)	N
0 Anti-stereotype	21.00	39.83	27.50	359
1 Stereotype	30.15	70.85	42.30	343
2 Neutral: no target	100.00	47.75	64.64	400
3 Neutral: target	0.00	0.00	0.00	372
4 Bias	10.00	2.27	3.70	264

Accuracy 33.54%

Macro F1 27.63%

Qwen LoRA · metadata / standalone

Class	P (%)	R (%)	F1 (%)	N
0 Anti-stereotype	87.07	91.92	89.43	359
1 Stereotype	92.19	86.01	88.99	343
2 Neutral: no target	100.00	100.00	100.00	400
3 Neutral: target	98.93	99.73	99.33	372
4 Bias	100.00	100.00	100.00	264

Accuracy 95.51%

Macro F1 95.55%

Complete test split: 1,738 examples. P = precision; R = recall; N = gold support. All five labels are scored.

Appendix · Classification Reports

Label-only · Full per-class precision, recall and F1

Qwen LoRA · metadata / train eval

Class	P (%)	R (%)	F1 (%)	N
0 Anti-stereotype	87.04	91.64	89.28	359
1 Stereotype	91.90	86.01	88.86	343
2 Neutral: no target	100.00	100.00	100.00	400
3 Neutral: target	98.93	99.73	99.33	372
4 Bias	100.00	100.00	100.00	264

Accuracy 95.45%

Macro F1 95.49%

ALBERT-base-v2

Class	P (%)	R (%)	F1 (%)	N
0 Anti-stereotype	76.79	71.87	74.24	359
1 Stereotype	72.27	79.01	75.49	343
2 Neutral: no target	98.21	96.00	97.09	400
3 Neutral: target	97.07	98.12	97.59	372
4 Bias	97.31	95.83	96.56	264

Accuracy 88.09%

Macro F1 88.20%

Complete test split: 1,738 examples. P = precision; R = recall; N = gold support. All five labels are scored.

Appendix · Classification Reports

Label-only · Full per-class precision, recall and F1

ModernBERT-base · original

Class	P (%)	R (%)	F1 (%)	N
0 Anti-stereotype	80.77	76.04	78.34	359
1 Stereotype	76.55	82.80	79.55	343
2 Neutral: no target	98.69	94.50	96.55	400
3 Neutral: target	96.02	97.31	96.66	372
4 Bias	95.17	96.97	96.06	264

Accuracy 89.36%

Macro F1 89.43%

ModernBERT-base · retuned

Class	P (%)	R (%)	F1 (%)	N
0 Anti-stereotype	85.51	82.17	83.81	359
1 Stereotype	82.17	86.01	84.05	343
2 Neutral: no target	99.48	96.00	97.71	400
3 Neutral: target	96.34	98.92	97.61	372
4 Bias	97.37	98.11	97.74	264

Accuracy 92.12%

Macro F1 92.18%

Complete test split: 1,738 examples. P = precision; R = recall; N = gold support. All five labels are scored.

Appendix · Classification Reports

Label-only · Full per-class precision, recall and F1

ModernBERT-large

Class	P (%)	R (%)	F1 (%)	N
0 Anti-stereotype	86.72	85.52	86.12	359
1 Stereotype	85.20	88.92	87.02	343
2 Neutral: no target	99.49	96.75	98.10	400
3 Neutral: target	97.60	98.39	97.99	372
4 Bias	98.09	97.35	97.72	264

Accuracy 93.33%

Macro F1 93.39%

Complete test split: 1,738 examples. P = precision; R = recall; N = gold support. All five labels are scored.

Appendix · Classification Reports

JSON category · Five normalized classes

Base JSON

Class	P (%)	R (%)	F1 (%)	N
gender	50.34	94.66	65.72	393
race	94.17	37.29	53.43	303
profession	73.68	4.61	8.68	607
religion	46.67	20.00	28.00	35
neutral	46.37	95.75	62.48	400

Accuracy 51.96%

Macro F1 43.66%

JSON LoRA · both evaluations

Class	P (%)	R (%)	F1 (%)	N
gender	98.98	98.98	98.98	393
race	100.00	100.00	100.00	303
profession	99.51	99.84	99.67	607
religion	100.00	100.00	100.00	35
neutral	98.99	98.50	98.75	400

Accuracy 99.37%

Macro F1 99.48%

Category/target predictions are identical in the two fine-tuned JSON evaluations; both sources are retained.

Appendix · Classification Reports

JSON label · All three full-test evaluations

Base JSON

Class	P (%)	R (%)	F1 (%)	N
0 Anti-stereotype	100.00	1.67	3.29	359
1 Stereotype	19.57	18.66	19.10	343
2 Neutral: no target	54.73	88.25	67.56	400
3 Neutral: target	44.00	17.74	25.29	372
4 Bias	18.03	41.67	25.17	264

Accuracy 34.46%

Macro F1 28.08%

JSON LoRA · standalone

Class	P (%)	R (%)	F1 (%)	N
0 Anti-stereotype	88.60	84.40	86.45	359
1 Stereotype	85.12	90.09	87.54	343
2 Neutral: no target	98.99	98.50	98.75	400
3 Neutral: target	96.58	98.66	97.61	372
4 Bias	99.61	96.21	97.88	264

Accuracy 93.61%

Macro F1 93.64%

Only the label field is scored here. Training-time and standalone JSON evaluations differ on one label.

Appendix · Classification Reports

JSON label · All three full-test evaluations

JSON LoRA · training-time test

Class	P (%)	R (%)	F1 (%)	N
0 Anti-stereotype	88.56	84.12	86.29	359
1 Stereotype	84.89	90.09	87.41	343
2 Neutral: no target	98.99	98.50	98.75	400
3 Neutral: target	96.58	98.66	97.61	372
4 Bias	99.61	96.21	97.88	264

Accuracy 93.56%

Macro F1 93.59%

Only the label field is scored here. Training-time and standalone JSON evaluations differ on one label.

Appendix · Target Classification

Base JSON · Exact target strings (1 / 3)

Gold target	P (%)	R (%)	F1 (%)	N
Afghanistan	40.00	18.18	25.00	11
Afghanistanis	100.00	27.27	42.86	22
African	66.67	33.33	44.44	6
Africans	100.00	87.50	93.33	16
Arab	100.00	20.00	33.33	10
Arabs	87.50	93.33	90.32	15
Asexual	0.00	0.00	0.00	17
Bartenders	100.00	14.75	25.71	61
Bengali	100.00	100.00	100.00	2
Bengalis	100.00	100.00	100.00	9
Bisexual	0.00	0.00	0.00	25
Brahmin	75.00	75.00	75.00	4
Brahmins	100.00	88.89	94.12	9
Butchers	100.00	56.52	72.22	23
Chemists	100.00	5.00	9.52	20

Gold target	P (%)	R (%)	F1 (%)	N
Commanders	100.00	60.00	75.00	20
Crimean	0.00	0.00	0.00	7
Crimeans	91.67	36.67	52.38	30
Engineers	100.00	20.83	34.48	24
Entrepreneurs	100.00	7.69	14.29	26
Ethiopian	66.67	22.22	33.33	9
Ethiopians	94.44	47.22	62.96	36
Gay	100.00	3.23	6.25	31
Gentlemen	100.00	4.55	8.70	22
Grandfathers	100.00	17.65	30.00	34
Guitarists	100.00	15.79	27.27	38
Hispanic	31.58	40.00	35.29	15
Hispanics	92.86	37.14	53.06	35
Historians	100.00	8.70	16.00	23
Japanese	100.00	55.56	71.43	18

Exact target accuracy 21.69% · Macro F1 43.47% · Out-of-set predictions count as errors.

Appendix · Target Classification

Base JSON · Exact target strings (2 / 3)

Gold target	P (%)	R (%)	F1 (%)	N
Lesbian	0.00	0.00	0.00	12
Lgbtq	0.00	0.00	0.00	19
Males	100.00	35.71	52.63	42
Managers	0.00	0.00	0.00	16
Mathematicians	100.00	10.53	19.05	19
Mommies	100.00	91.67	95.65	12
Mothers	0.00	0.00	0.00	22
Muslim	100.00	100.00	100.00	1
Muslims	100.00	28.57	44.44	21
Nb	0.00	0.00	0.00	15
Neutral	0.00	0.00	0.00	400
Nurses	100.00	75.00	85.71	8
Pansexual	0.00	0.00	0.00	5
Performing_artists	0.00	0.00	0.00	41
Physicists	0.00	0.00	0.00	11

Gold target	P (%)	R (%)	F1 (%)	N
Politicians	0.00	0.00	0.00	21
Producers	100.00	2.22	4.35	45
Psychologists	0.00	0.00	0.00	24
Queer	0.00	0.00	0.00	10
Researchers	100.00	9.09	16.67	22
Russian	40.00	13.33	20.00	15
Russians	90.91	71.43	80.00	14
Schoolboys	100.00	80.00	88.89	15
Schoolgirls	100.00	57.14	72.73	14
Sisters	100.00	61.54	76.19	13
Software_developers	0.00	0.00	0.00	54
Spain	100.00	50.00	66.67	6
Spanish	88.89	66.67	76.19	12
Transgender	0.00	0.00	0.00	20
Yemen	0.00	0.00	0.00	2

Exact target accuracy 21.69% · Macro F1 43.47% · Out-of-set predictions count as errors.

Appendix · Target Classification

Base JSON · Exact target strings (3 / 3)

Gold target	P (%)	R (%)	F1 (%)	N
Yemenis	100.00	30.77	47.06	13
bartender	75.00	100.00	85.71	3
butcher	100.00	16.67	28.57	6
chemist	100.00	50.00	66.67	4
commander	100.00	25.00	40.00	8
engineer	100.00	25.00	40.00	4
entrepreneur	100.00	33.33	50.00	3
gentlemen	28.57	100.00	44.44	4
grandfather	66.67	66.67	66.67	3
guitarist	100.00	25.00	40.00	4
historian	0.00	0.00	0.00	2
male	71.43	100.00	83.33	5
manager	100.00	16.67	28.57	6
mathematician	0.00	0.00	0.00	1
mommy	100.00	83.33	90.91	6

Gold target	P (%)	R (%)	F1 (%)	N
mother	93.75	65.22	76.92	23
nurse	100.00	50.00	66.67	12
performing artist	75.00	50.00	60.00	6
physicist	100.00	46.15	63.16	13
politician	50.00	50.00	50.00	2
producer	100.00	20.00	33.33	10
psychologist	88.89	61.54	72.73	13
researcher	100.00	83.33	90.91	6
schoolboy	100.00	80.00	88.89	5
schoolgirl	92.86	100.00	96.30	13
sister	100.00	83.33	90.91	6
software developer	100.00	37.50	54.55	8

Exact target accuracy 21.69% · Macro F1 43.47% · Out-of-set predictions count as errors.

Appendix · Target Classification

JSON LoRA · both evaluations · Exact target strings (1 / 3)

Gold target	P (%)	R (%)	F1 (%)	N
Afghanistan	100.00	100.00	100.00	11
Afghanistanis	100.00	100.00	100.00	22
African	100.00	100.00	100.00	6
Africans	100.00	100.00	100.00	16
Arab	100.00	80.00	88.89	10
Arabs	88.24	100.00	93.75	15
Asexual	100.00	100.00	100.00	17
Bartenders	100.00	100.00	100.00	61
Bengali	100.00	50.00	66.67	2
Bengalis	90.00	100.00	94.74	9
Bisexual	100.00	100.00	100.00	25
Brahmin	100.00	100.00	100.00	4
Brahmins	100.00	100.00	100.00	9
Butchers	100.00	95.65	97.78	23
Chemists	100.00	100.00	100.00	20

Gold target	P (%)	R (%)	F1 (%)	N
Commanders	100.00	100.00	100.00	20
Crimean	100.00	85.71	92.31	7
Crimeans	96.77	100.00	98.36	30
Engineers	100.00	100.00	100.00	24
Entrepreneurs	100.00	100.00	100.00	26
Ethiopian	100.00	88.89	94.12	9
Ethiopians	97.30	100.00	98.63	36
Gay	100.00	100.00	100.00	31
Gentlemen	95.24	90.91	93.02	22
Grandfathers	100.00	100.00	100.00	34
Guitarists	100.00	100.00	100.00	38
Hispanic	100.00	100.00	100.00	15
Hispanics	100.00	100.00	100.00	35
Historians	95.65	95.65	95.65	23
Japanese	100.00	94.44	97.14	18

Exact target accuracy 98.16% · Macro F1 97.51% · Out-of-set predictions count as errors.

Appendix · Target Classification

JSON LoRA · both evaluations · Exact target strings (2 / 3)

Gold target	P (%)	R (%)	F1 (%)	N
Lesbian	100.00	100.00	100.00	12
Lgbtq	100.00	100.00	100.00	19
Males	100.00	100.00	100.00	42
Managers	93.75	93.75	93.75	16
Mathematicians	90.48	100.00	95.00	19
Mommies	100.00	100.00	100.00	12
Mothers	95.65	100.00	97.78	22
Muslim	100.00	100.00	100.00	1
Muslims	100.00	100.00	100.00	21
Nb	100.00	93.33	96.55	15
Neutral	98.99	98.50	98.75	400
Nurses	100.00	100.00	100.00	8
Pansexual	100.00	100.00	100.00	5
Performing_artists	95.24	97.56	96.39	41
Physicists	100.00	100.00	100.00	11

Gold target	P (%)	R (%)	F1 (%)	N
Politicians	100.00	100.00	100.00	21
Producers	100.00	100.00	100.00	45
Psychologists	100.00	100.00	100.00	24
Queer	100.00	100.00	100.00	10
Researchers	95.24	90.91	93.02	22
Russian	100.00	93.33	96.55	15
Russians	93.33	100.00	96.55	14
Schoolboys	100.00	100.00	100.00	15
Schoolgirls	100.00	100.00	100.00	14
Sisters	100.00	76.92	86.96	13
Software_developers	96.43	100.00	98.18	54
Spain	100.00	83.33	90.91	6
Spanish	92.31	100.00	96.00	12
Transgender	100.00	100.00	100.00	20
Yemen	100.00	100.00	100.00	2

Exact target accuracy 98.16% · Macro F1 97.51% · Out-of-set predictions count as errors.

Appendix · Target Classification

JSON LoRA · both evaluations · Exact target strings (3 / 3)

Gold target	P (%)	R (%)	F1 (%)	N
Yemenis	100.00	100.00	100.00	13
bartender	100.00	100.00	100.00	3
butcher	100.00	100.00	100.00	6
chemist	100.00	100.00	100.00	4
commander	100.00	87.50	93.33	8
engineer	100.00	100.00	100.00	4
entrepreneur	100.00	100.00	100.00	3
gentlemen	100.00	75.00	85.71	4
grandfather	75.00	100.00	85.71	3
guitarist	100.00	100.00	100.00	4
historian	100.00	100.00	100.00	2
male	83.33	100.00	90.91	5
manager	100.00	100.00	100.00	6
mathematician	100.00	100.00	100.00	1
mommy	100.00	100.00	100.00	6

Gold target	P (%)	R (%)	F1 (%)	N
mother	100.00	95.65	97.78	23
nurse	100.00	91.67	95.65	12
performing artist	100.00	83.33	90.91	6
physicist	100.00	92.31	96.00	13
politician	100.00	100.00	100.00	2
producer	100.00	100.00	100.00	10
psychologist	100.00	100.00	100.00	13
researcher	100.00	100.00	100.00	6
schoolboy	100.00	100.00	100.00	5
schoolgirl	100.00	100.00	100.00	13
sister	100.00	100.00	100.00	6
software developer	100.00	100.00	100.00	8

Exact target accuracy 98.16% · Macro F1 97.51% · Out-of-set predictions count as errors.

Appendix · Report Coverage

Complete reports and diagnostic subsets are kept distinct

Included in the test appendix

- 9 label-only test evaluations
- 3 JSON test evaluations
- All 5 label/category classes
- All 87 exact target classes

Identical category/target reports are shared across the two LoRA JSON evaluations.

Companion TXT files

Validation class reports and partial diagnostics are labeled separately.

The unfinished 410-row benchmark is not a full-test result.

Saved examples and metrics are preserved without rerunning models.

Companions: all_label_only_classification_reports.txt and all_category_label_target_classification_reports.txt